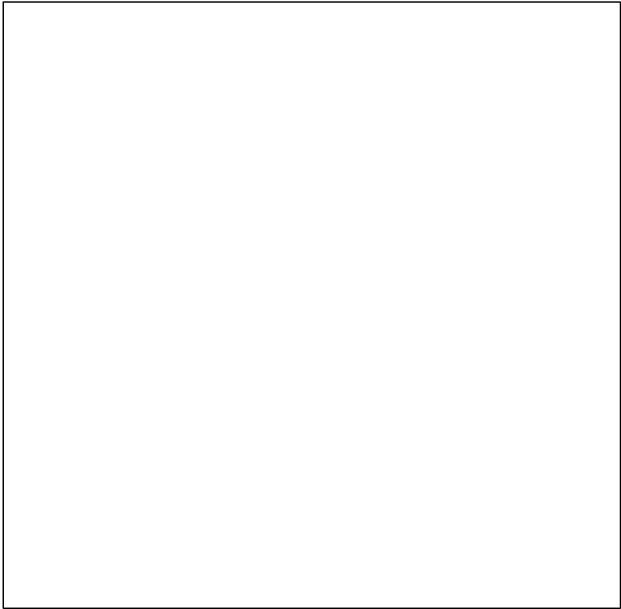


Schematic: Top-Level Hierarchy

Overview of the five functional sheets of the design.

ESP32



File: ESP32.kicad_sch

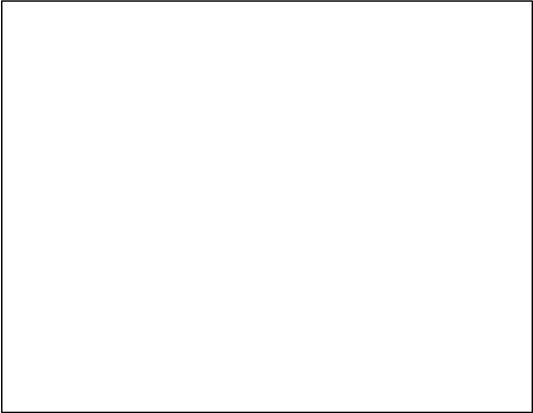
POWER SUPPLY



File: POWER SUPPLY.kicad_sch
Switch Input Section

GNDD

RELAY SECTION



File: RELAY SECTION.kicad_sch

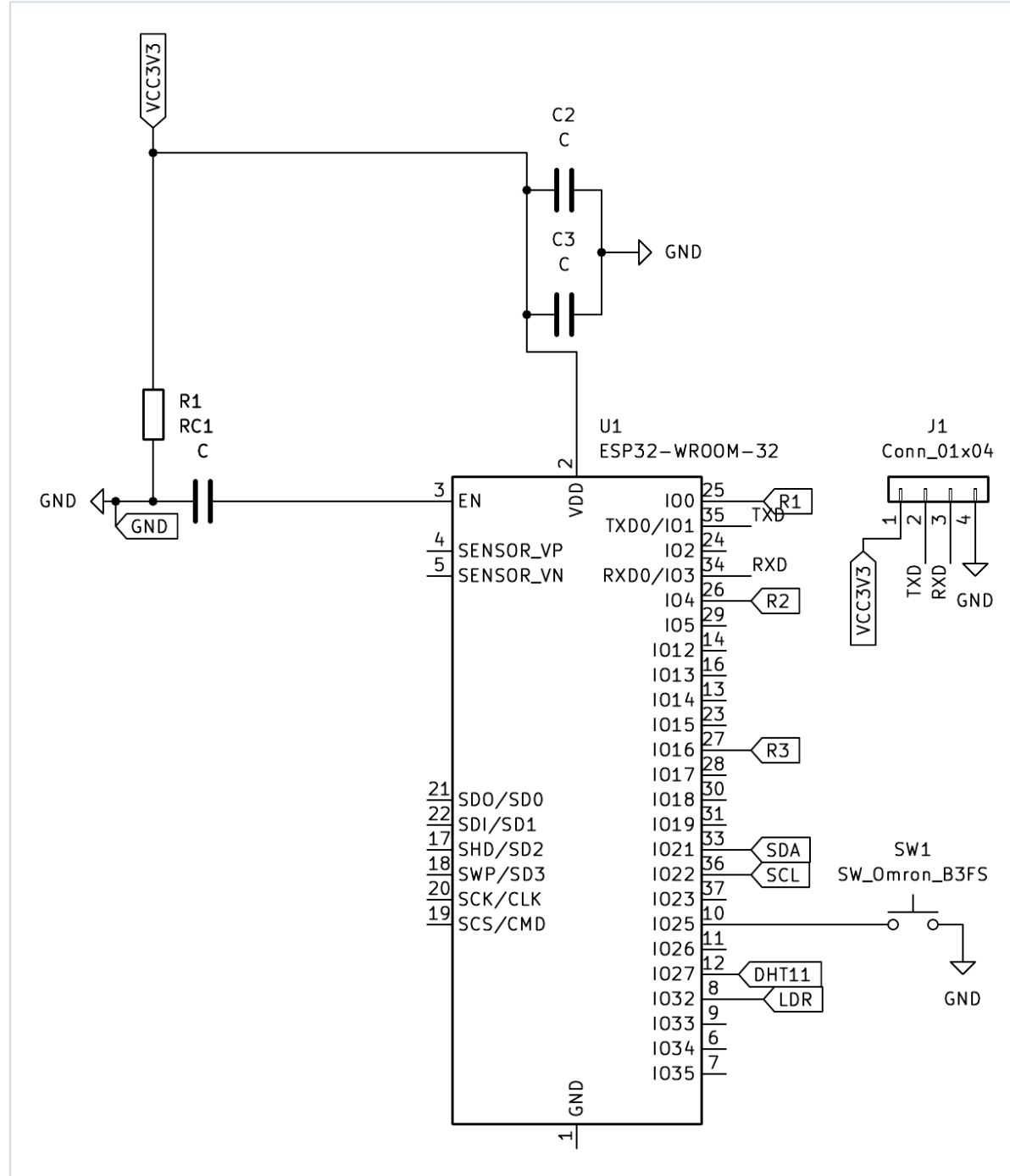
SENSOR & LCD



File: SENSOR.kicad_sch

Schematic: ESP32 Controller

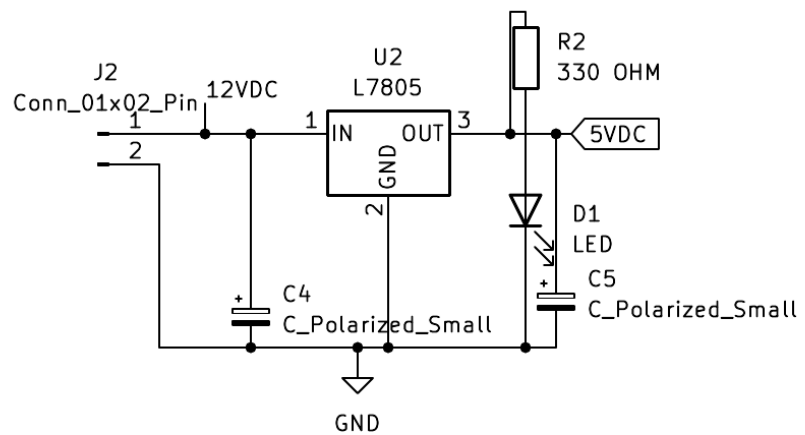
ESP32-WROOM-32 module with UART header and push-button.



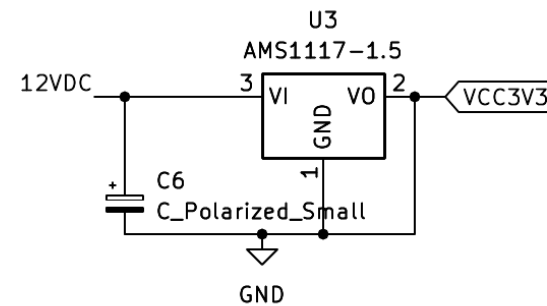
Schematic: Power Supply

12 V input regulated to 5 V (L7805) and 3.3 V (AMS1117).

5V POWER SUPPLY



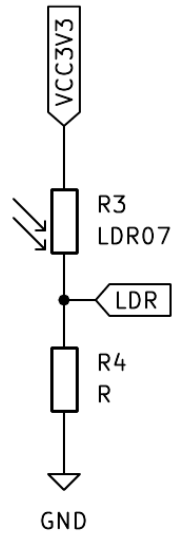
3.3V POWER SUPPLY



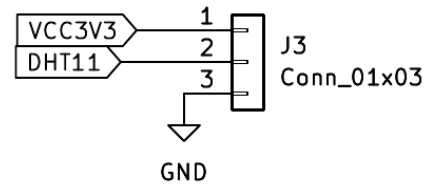
Schematic: Sensor & LCD

LDR, DHT11 and I2C LCD interface.

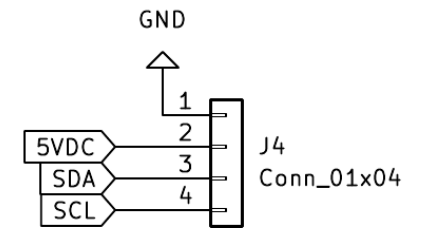
LDR



DHT11

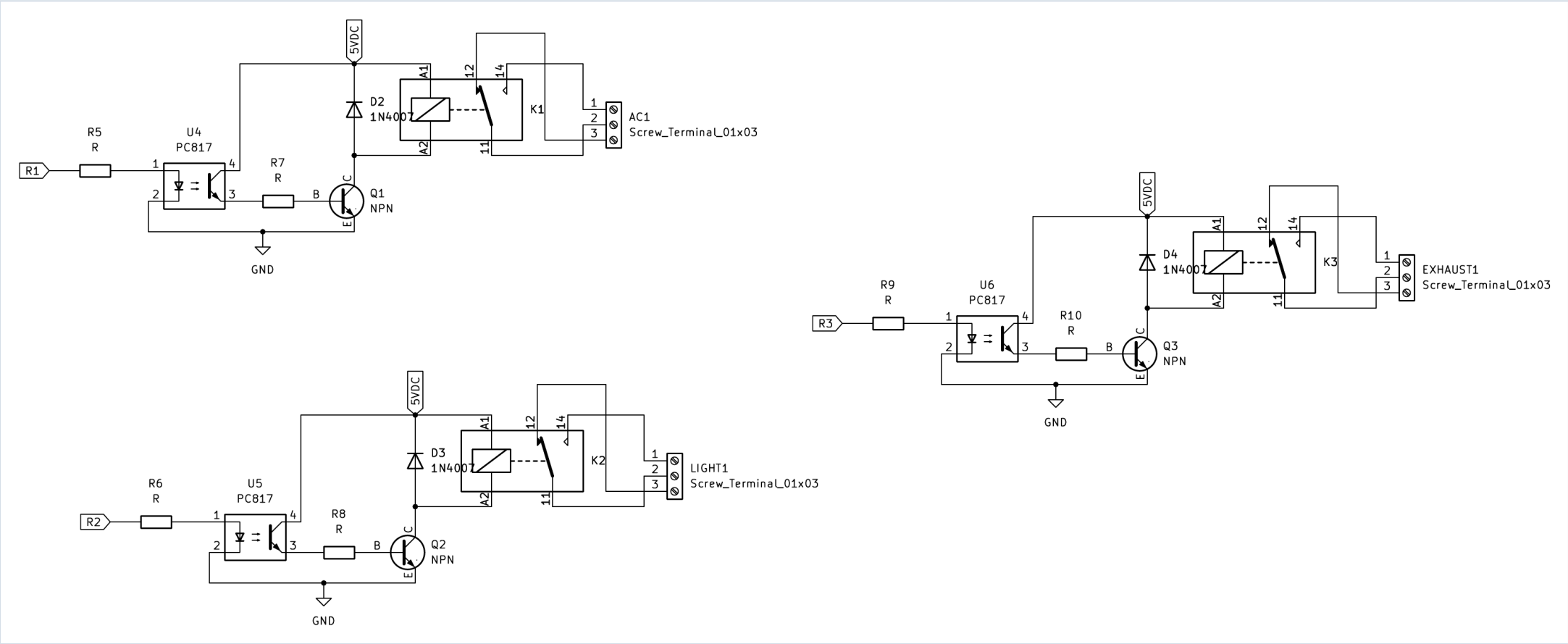


LCD



Schematic: Relay Section

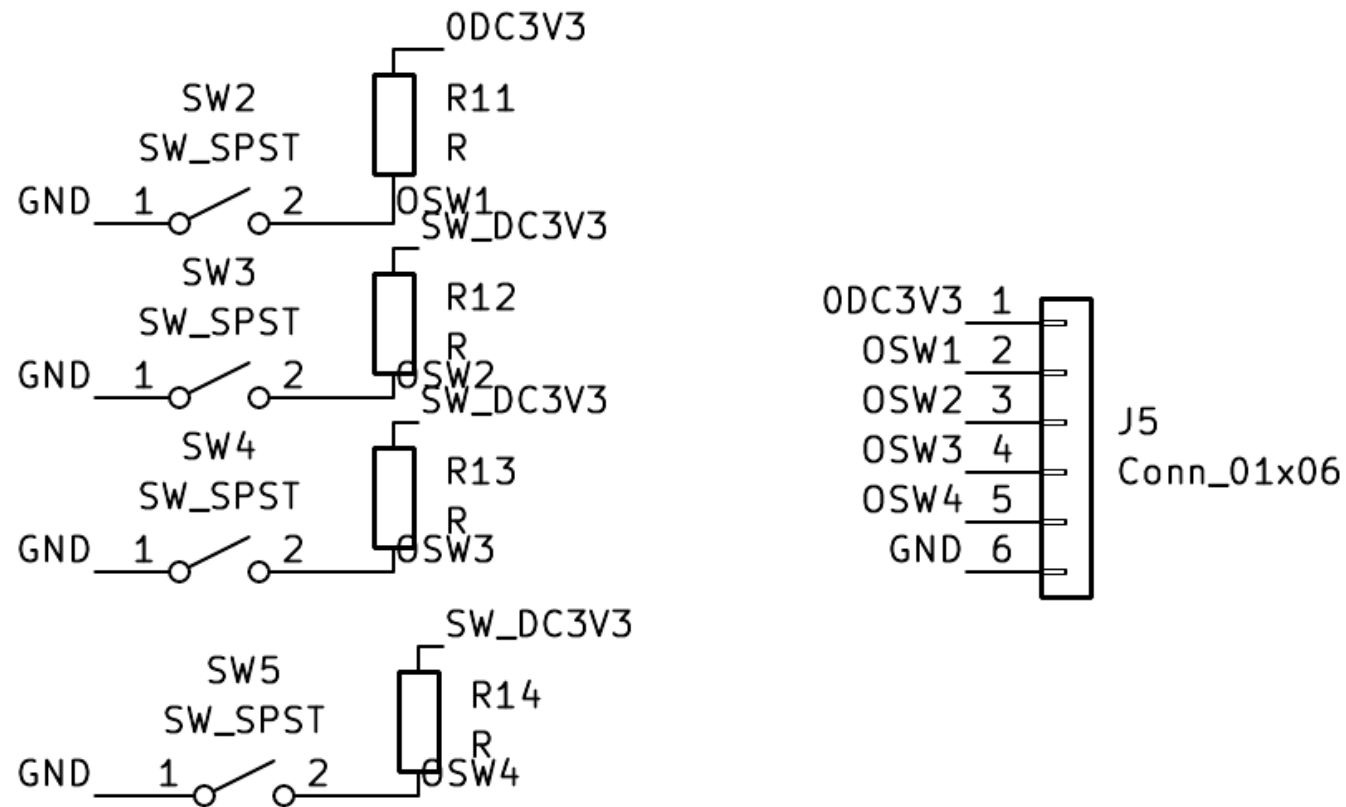
Three opto-isolated relay channels: Light, AC and Exhaust.



Schematic: Switch Input Section

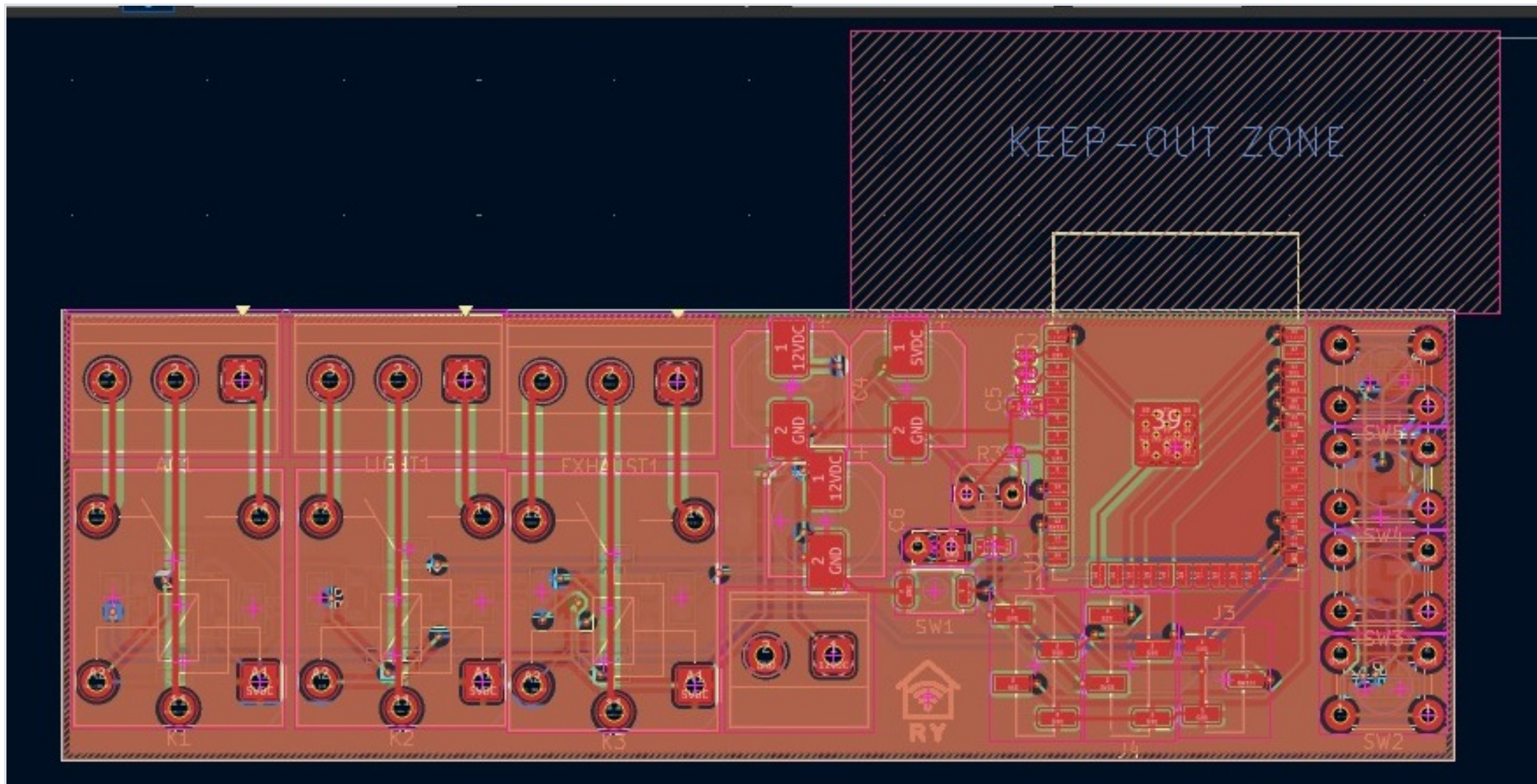
Four push-button inputs routed to the ESP32 through header J5.

SWITCH INPUT SECTION



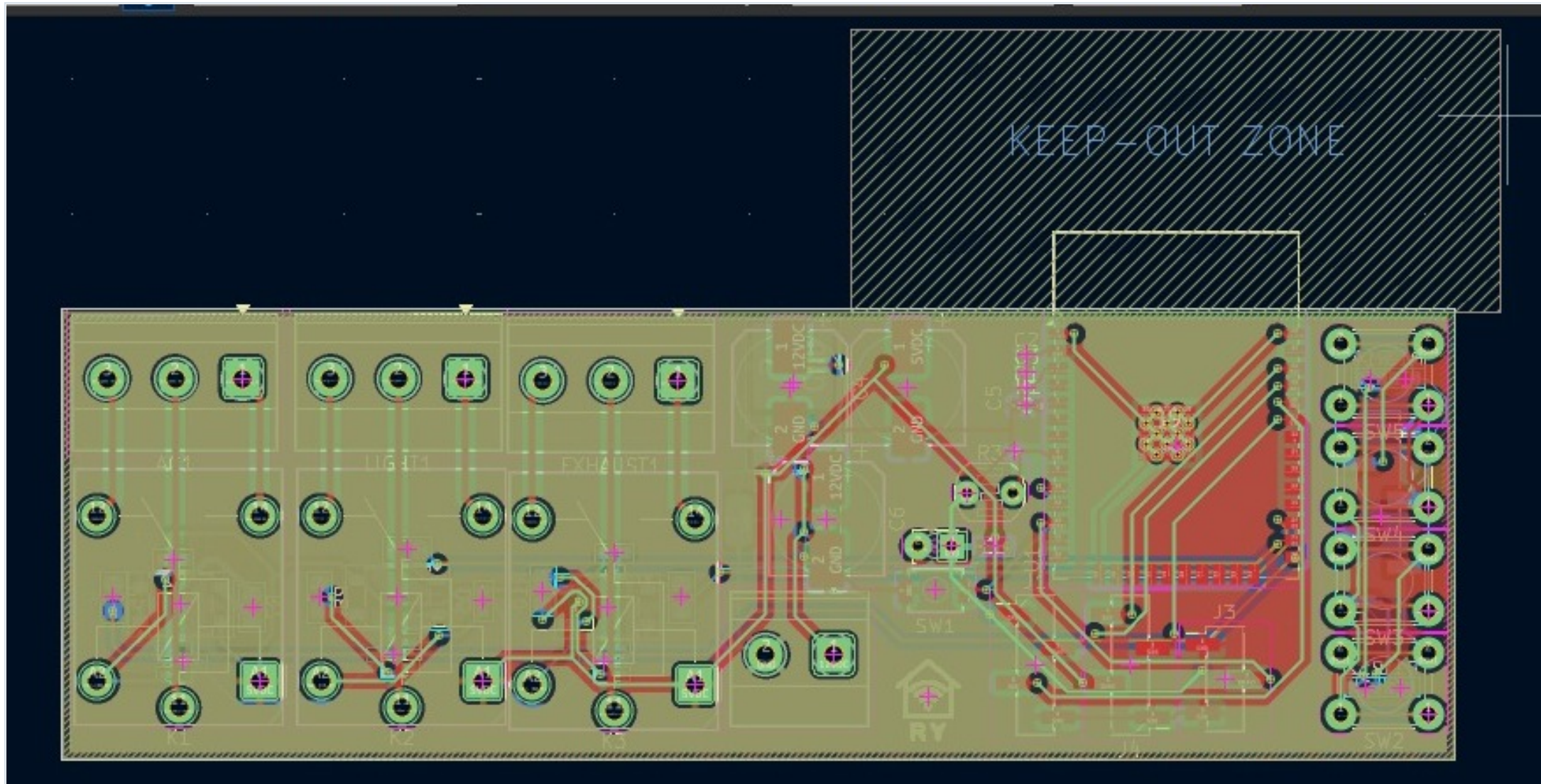
PCB Layout: Top Layer (F.Cu)

4-layer board, fully routed with 0 unrouted nets.



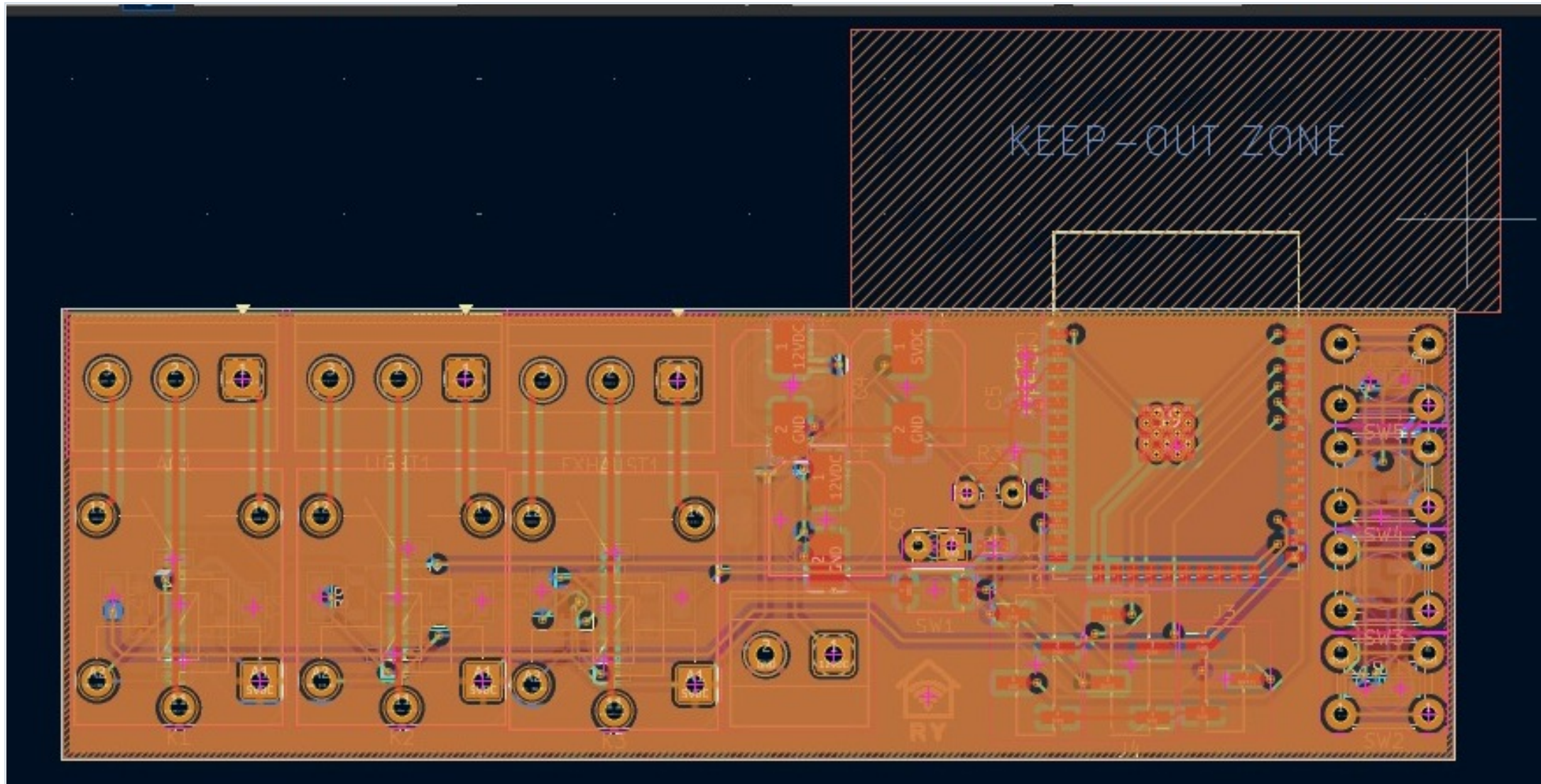
PCB Layout: Inner Layer 1 (In1.Cu)

Internal copper layer.



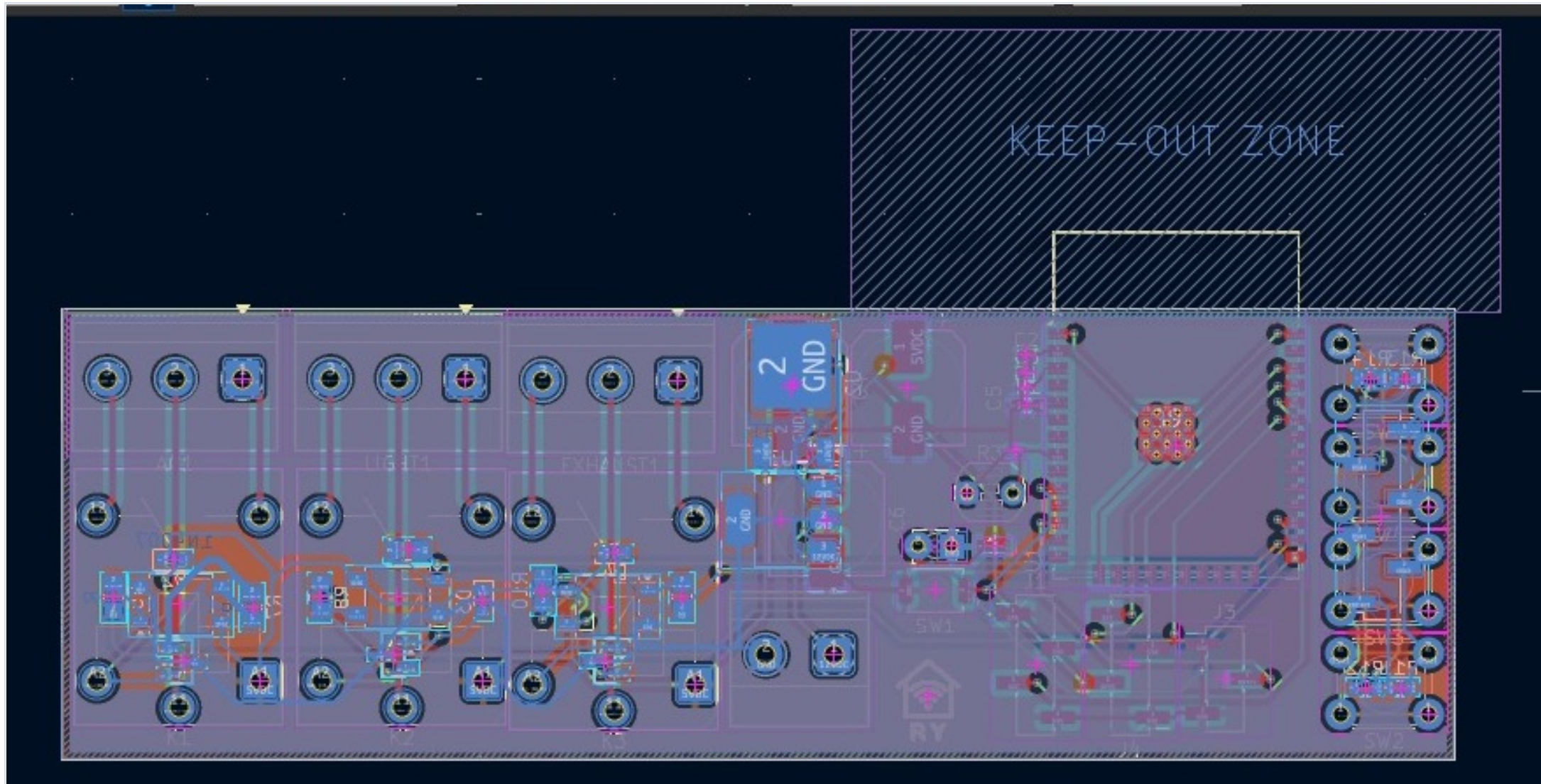
PCB Layout: Inner Layer 2 (In2.Cu)

Internal copper layer.



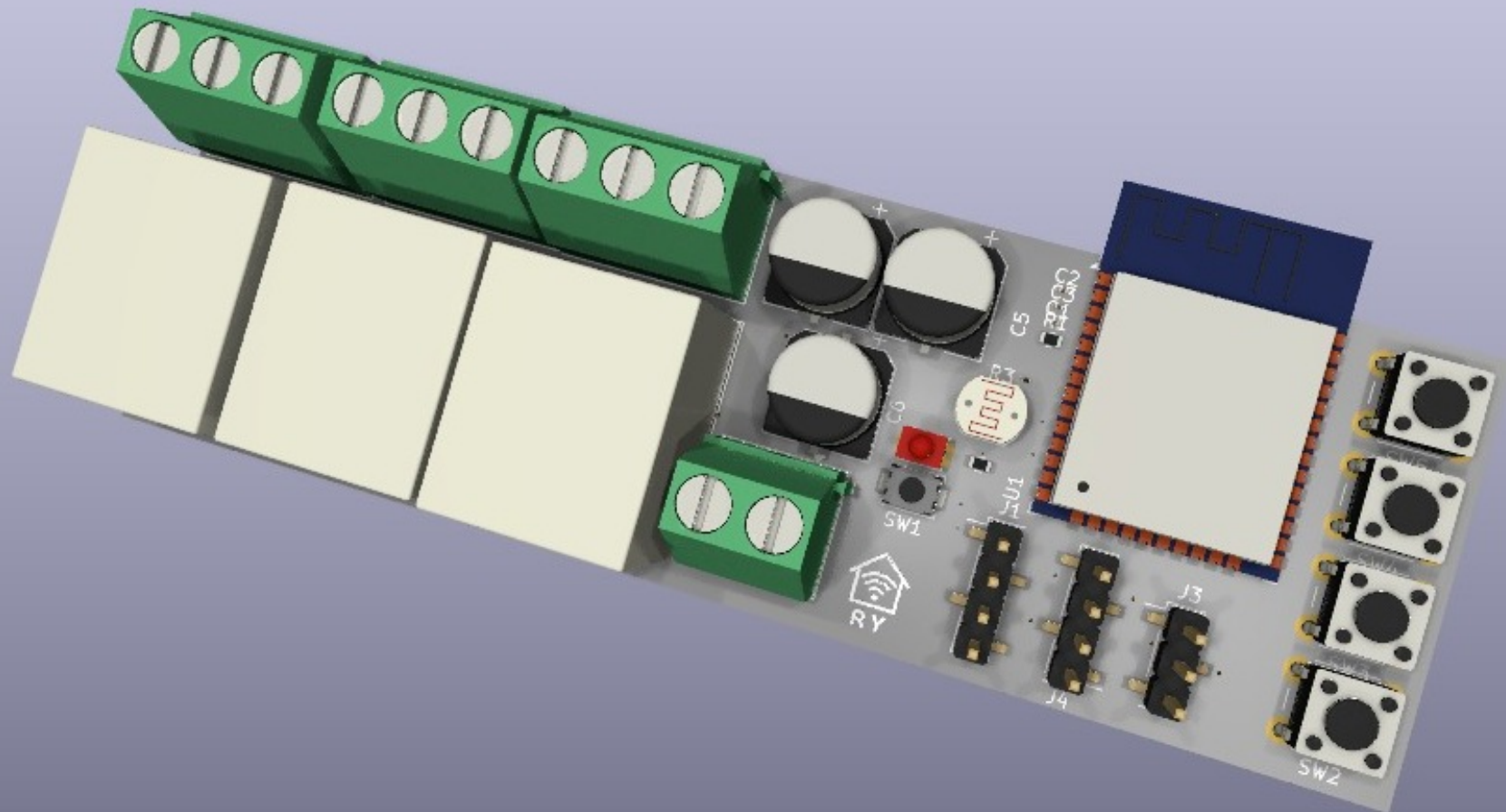
PCB Layout: Bottom Layer (B.Cu)

SMD components and remaining routing on the bottom side.



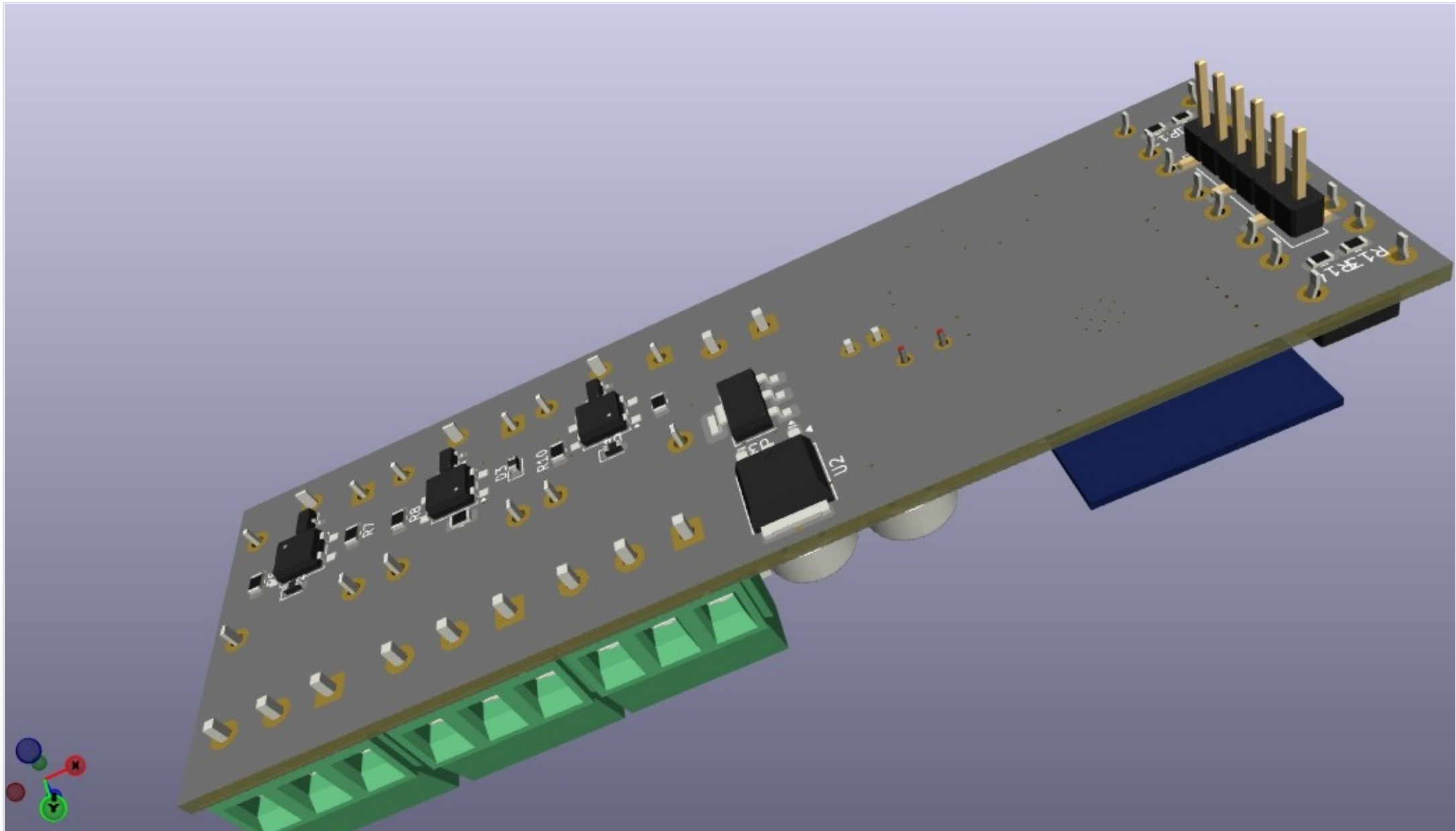
3D View: Top Side

Relays, screw terminals, ESP32 module and push-buttons.



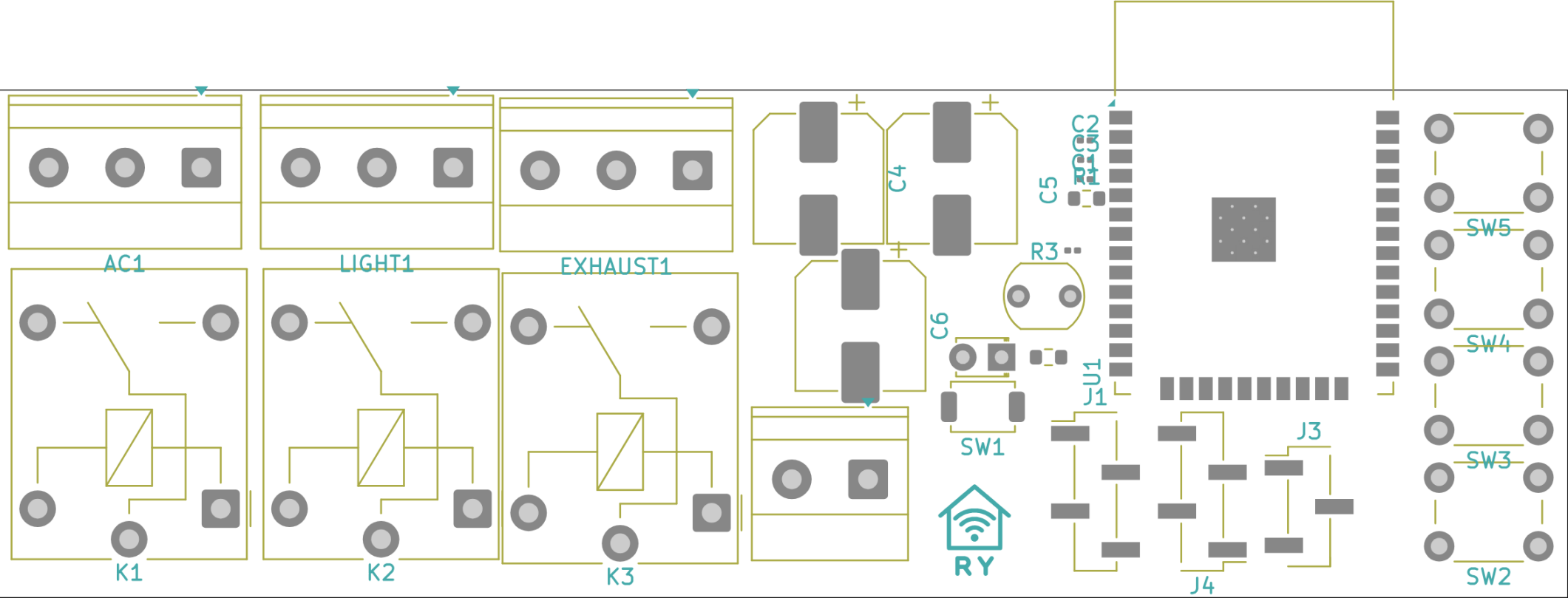
3D View: Bottom Side

SMD components and headers on the bottom side.



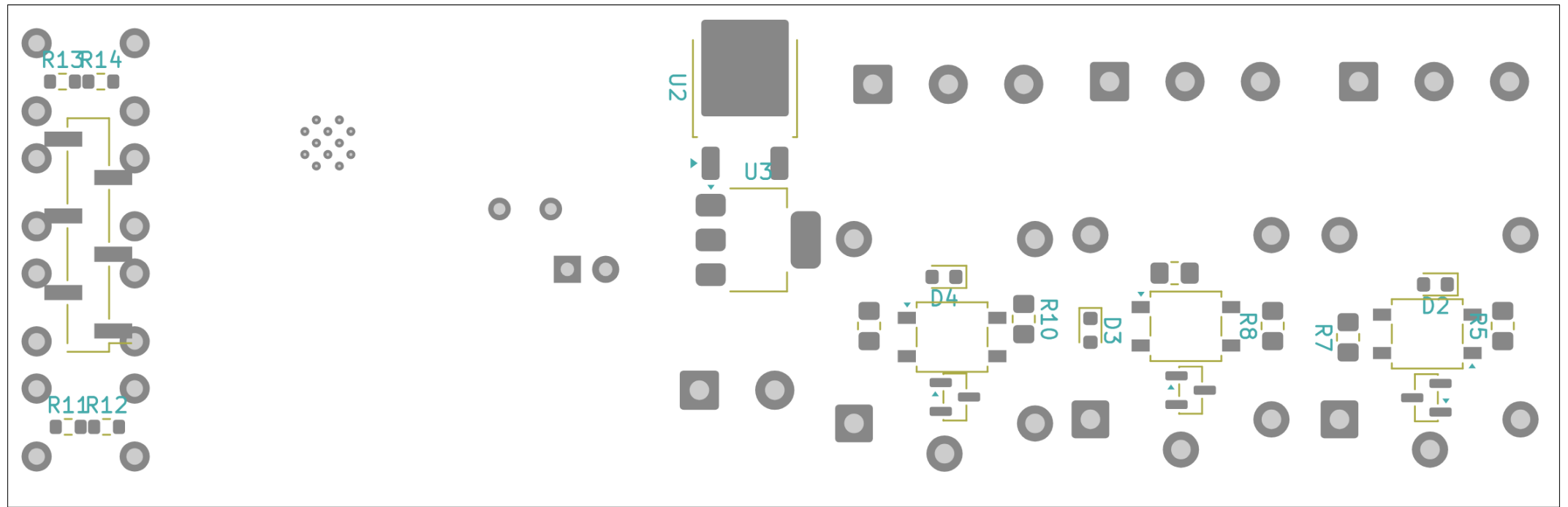
PCB Assembly View: Front

Top side component placement and silkscreen.



PCB Assembly View: Back

Bottom side component placement and pads.



Bill of Materials (BOM)

Home Automation PCB: 49 components, 23 unique line items.

#	Designator	Qty	Value	Footprint	Description
1	C1, C2, C3	3	100 nF *	C_0201_0603Metric	Ceramic capacitor
2	C4, C5, C6	3	100 uF, 25 V *	CP_Elec_8x6.2	Polarized electrolytic capacitor
3	R1, R11, R12, R13, R14	5	10 k ohm *	R_0603_1608Metric	Resistor
4	R2	1	330 ohm	R_0603_1608Metric	Resistor (power LED)
5	R3	1	LDR07	R_LDR_5.1x4.3mm_P3.4mm_Vertical	Light dependent resistor
6	R4	1	10 k ohm *	R_0201_0603Metric	Resistor (LDR divider)
7	R5, R6, R9	3	330 ohm *	R_0805_2012Metric_Pad1.20x1.40mm_HandSolder	Resistor (optocoupler input)
8	R7, R8, R10	3	1 k ohm *	R_0805_2012Metric_Pad1.20x1.40mm_HandSolder	Resistor (transistor base)
9	D1	1	LED	LED_D1.8mm_W3.3mm_H2.4mm	Power indicator LED
10	D2, D3, D4	3	1N4007	D_0603_1608Metric	Relay flyback diode
11	Q1, Q2, Q3	3	NPN	SOT-23	NPN transistor (relay driver)
12	U1	1	ESP32-WROOM-32	ESP32-WROOM-32	Wi-Fi/Bluetooth microcontroller module
13	U2	1	L7805	TO-252-2	5 V voltage regulator
14	U3	1	AMS1117	SOT-223-3_TabPin2	3.3 V LDO regulator
15	U4, U5, U6	3	PC817	SO-4_4.4x4.3mm_P2.54mm	Optocoupler
16	K1, K2, K3	3	FTR-LYCA005x	Relay_SPDT_Finder_36.11	5 V SPDT relay
17	SW1	1	SW_Omron_B3FS	SW_SPST_TS-1088-xR020	Tactile switch
18	SW2, SW3, SW4, SW5	4	SW_SPST	SW_PUSH_6mm	Push button
19	J1, J4	2	Conn_01x04	PinHeader_1x04_P2.54mm_Vertical_SMD_Pin1Left	4-pin header (UART / LCD)
20	J3	1	Conn_01x03	PinHeader_1x03_P2.54mm_Vertical_SMD_Pin1Left	3-pin header (DHT11)
21	J5	1	Conn_01x06	PinHeader_1x06_P2.54mm_Vertical_SMD_Pin1Left	6-pin header (switch inputs)
22	J2	1	Conn_01x02	TerminalBlock_Phoenix_MKDS-1,5-2_1x02_P5.00mm_Horizontal	2-pin screw terminal (12 V input)
23	AC1, LIGHT1, EXHAUST1	3	Screw_Terminal_01x03	TerminalBlock_Phoenix_MKDS-1,5-3_1x03_P5.00mm_Horizontal	3-pin screw terminal (load output)

* Value selected from the circuit function; confirm before ordering.